

PBI-Assisted Quality Classification and Criterion-Diversified Infill Selection for Expensive Many-Objective Optimization

[TODO: author list]

Department, University, Street, City, 000000, Province, Country

Abstract

[TODO: Use the challenge–contribution structure. State the two concrete gaps: REMO forms training groups using a small set of representatives selected through RSEA’s two-dimensional radial projection. These representatives may not fully reflect the current nondominated distribution in the original objective space, and relation scores alone leave the use of prediction ambiguity and indicator information in candidate selection unspecified. Introduce PACDIS with PBI-assisted quality classification (PAQC) and criterion-diversified infill selection (CDIS). PAQC constructs adaptive directions directly from the current nondominated objective vectors and uses PBI-based directional quality information to supplement representative-based positive-group construction. CDIS allocates expensive evaluations through ambiguity-oriented and indicator-oriented selection criteria. Present the two mechanisms as complementary, with problem-dependent effects in the component comparisons. Lead the results with the strongest verified algorithm-level IGD⁺ comparison in the tested many-objective settings. Then give one concise mechanism finding about group composition or candidate-batch utility. Keep detailed overlap statistics and componentwise tests in the experimental section.]

Keywords: expensive many-objective optimization, surrogate-assisted evolutionary algorithm, relation learning, penalty-based boundary intersection, infill selection

1. Introduction

Expensive many-objective optimization problems (MaOPs) arise in many real-world applications where each objective evaluation is computationally or economically prohibitive, limiting the number of real function evaluations to a small budget. Conventional many-objective evolutionary algorithms (MaOEAs) typically require tens of thousands of evaluations, making them impractical in such scenarios. Surrogate-assisted evolutionary algorithms (SAEAs) alleviate this burden by approximating objective functions or other performance indicators [11, 9, 7]. However, building and maintaining accurate objective-wise regression models becomes increasingly challenging as the number of objectives grows, especially when only a limited number of expensive samples are available. Relation learning offers an alternative: instead of recovering absolute objective values, it learns the relative quality of pairs of solutions [2]. This paradigm has shown promise in expensive multi-objective optimization. However, its effectiveness is influenced not only by the relation model itself, but also by how supervision is constructed and how its predictions are converted into expensive evaluations.

REMO [2] constructs its training groups from a small set of representative solutions selected by the radial-space mechanism of RSEA [3]. RSEA projects high-dimensional objective vectors into a two-dimensional radial space and uses grid division for efficient selection. Although this reduction enables efficient representative selection, the resulting representative set is a sparse summary of the population. Consequently, relying ex-

clusively on these representatives to form training groups may underuse the directional information provided by the current nondominated solutions in the original objective space. To address this limitation, we propose PBI-assisted quality classification (PAQC). PAQC derives nondominated-derived directional quality information directly from the current nondominated objective vectors using PBI, without relying on radial-grid representative selection. It then fuses this directional quality information with representative-based binary labels to enrich the information used to construct the positive training group. In this way, PAQC enriches the supervision signal for relation learning while retaining REMO’s three-class relation-learning scheme. A stage-dependent fusion strategy is further used to adjust the influence of the two signals over the course of optimization.

Candidate selection poses a second problem: relation scores order candidates, but that ordering alone does not specify how to use classification ambiguity or a separate quality indicator when allocating a small evaluation batch. Criterion-diversified infill selection (CDIS) addresses this evaluation-allocation interface through two alternative criteria. The ambiguity-oriented mode ranks candidates by relation quality and classification ambiguity after quality screening. The indicator-oriented mode reranks a relation-screened subset using a separately learned quality indicator. Each iteration activates one criterion to select a batch within the remaining evaluation budget.

We propose PACDIS, a surrogate-assisted many-objective evolutionary algorithm that modifies two interfaces of relation-based expensive optimization: supervision construction and

expensive-evaluation allocation. PAQC determines what the relation classifier learns from evaluated solutions; CDIS determines how its predictions guide new expensive evaluations. The main contributions of this work are threefold. First, PAQC supplements representative-based grouping with nondominated-derived directional quality information and combines the two signals to construct positive training groups. Second, CDIS uses relation quality, classification ambiguity, and indicator information through two alternative infill criteria to allocate a limited expensive-evaluation budget. These mechanisms address different stages of the relation-learning framework. Third, experiments on DTLZ and WFG benchmarks with 10, 15, and 20 objectives compare PACDIS with six surrogate-assisted algorithms, while deletion controls examine the respective roles of PAQC and CDIS. PACDIS shows competitive performance on the tested settings, with consistent mean IGD⁺ advantages on WFG2 and WFG7 across the three objective counts; HES-EA remains stronger on DTLZ5 and DTLZ7. The reported component comparisons suggest complementary, problem-dependent benefits. Group-quality and candidate-quality diagnostics provide the planned mechanism analyses.

The remainder of this paper is organized as follows. Section 2 reviews related work on surrogate-assisted expensive optimization, relation learning, and candidate selection. Section 3 details the proposed PACDIS, including PAQC, relation learning on the constructed groups, and CDIS. Section 4 presents the experimental studies and discussion. Section 5 concludes the paper and outlines future work.

2. Related Work

2.1. Surrogate-Assisted Expensive Multi-/Many-Objective Optimization

[TODO: Kriging/RBF-based SAEAs, their scalability limit in M , and why regression on every objective becomes unattractive when M grows.]

2.2. Relation Learning for Expensive Optimization

[TODO: REMO and the pairwise-comparison SAEA line; compare how each method forms training groups or pairwise labels, then position the present work as group construction that supplements directions induced by radially selected representatives with directions obtained directly from the current nondominated set. Explain how directional quality information and the original binary labels jointly determine training-group membership.]

2.3. Quality Assessment and Candidate Selection

[TODO: Indicator-based assessment (SDE, L_p shape estimation, PIEA), radial representative selection (RSEA), and infill/candidate selection rules.]

RSEA uses radial projection and adaptive grids for selection, with both crowding and convergence considered to address the effects of curvature information loss [3]. PAQC retains this

representative-selection mechanism and supplements its grouping signal with a separate assessment in the original objective space. The distinction concerns the source of directional information, rather than the binary form of the original labels alone.

3. The Proposed PACDIS

Relation-based surrogate-assisted evolutionary algorithms replace multi-output regression by a classifier that predicts the relative quality of a pair of solutions [2]. Two interfaces then decide what such an algorithm can learn and how much its predictions cost: the rule that turns the evaluated population into supervision labels, and the rule that turns the surrogate scores of a large candidate pool into a small batch of expensive evaluations. We propose PACDIS, a surrogate-assisted many-objective evolutionary algorithm that modifies supervision construction through PAQC and expensive-evaluation allocation through CDIS. *PBI-assisted quality classification* (PAQC) augments the representative-based grouping of REMO [2] with PBI-based quality information from directions constructed directly from the current nondominated set. This additional source enriches the distribution information available for relation-group construction. PAQC fuses the two signals to determine which solutions enter the positive training group. *Criterion-diversified infill selection* (CDIS) retains relation-guided candidate generation but uses either indicator-based reranking or ambiguity-rewarded ranking to choose the final expensive-evaluation batch. Established relation learning, indicator assessment and environmental selection modules provide the host algorithm; the two proposed mechanisms change how its training groups are formed and how its surrogate scores are used.

3.1. Overall Framework

We consider the expensive many-objective problem

$$\min_{\mathbf{x} \in \Omega} \mathbf{f}(\mathbf{x}) = [f_1(\mathbf{x}), \dots, f_M(\mathbf{x})], \quad \Omega \subseteq \mathbb{R}^D, \quad (1)$$

in which every evaluation of $\mathbf{f}$ is expensive, so the number of real function evaluations is limited to a budget $FE_{\max}$. Let FE be the number of evaluations consumed so far and $t = FE/FE_{\max} \in [0, 1]$ the evaluation progress, which controls the hybrid score in PAQC.

PACDIS evaluates a Latin hypercube design of N_{init} solutions, stores them in an archive $\mathcal{A}_{\text{rc}}$, and then repeats five steps until the budget is exhausted. (i) The current population $\mathcal{P}$ is divided into a positive group and a non-positive group by PAQC, which also returns the set $\mathcal{R}$ of representative solutions reused later as an additional mating pool. (ii) Ordered pairs are generated from the two groups and a three-class relation network is trained using the resulting ordered group labels. (iii) An indicator value is computed for each solution in $\mathcal{P}$ and an RBF-SVR model is fitted from decision vectors to those values. (iv) One of the two selection modes is drawn, a relation-guided inner search accumulates candidates under that mode, and a batch of at most $n_{\max}$ candidates is returned. (v) The batch is evaluated, appended to $\mathcal{A}_{\text{rc}}$, and the next population is obtained by

Algorithm 1 PACDIS

Input: problem $(\Omega, \mathbf{f})$, budget $FE_{\max}$, initial size N_{init} , population size N

Output: archive $\mathcal{A}_{\text{rc}}$ of evaluated solutions

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1:  $\mathcal{P} \leftarrow N_{\text{init}}$  solutions sampled by Latin hypercube design
2: Evaluate  $\mathcal{P}$ ;  $FE \leftarrow N_{\text{init}}$ 
3:  $\mathcal{A}_{\text{rc}} \leftarrow \mathcal{P}$ 
4: while  $FE < FE_{\max}$  do
5:    $t \leftarrow FE/FE_{\max}$ 
6:    $(C_1, C_2, \mathcal{R}) \leftarrow \text{PBIQUALITYCLASSIFICATION}(\mathcal{P}, t)$ 
7:   Build ordered relation pairs from  $C_1, C_2$  by (8)
8:   Train the relation surrogate  $\mathcal{M}_R$ 
9:   Train the indicator surrogate  $\mathcal{M}_I$  on  $\mathcal{P}$ 
10:  Draw an ambiguity-oriented or indicator-oriented mode  $m$ 
11:   $\mathcal{S} \leftarrow \text{CDIS}(\mathcal{P}, \mathcal{R}, \mathcal{M}_R, \mathcal{M}_I, m)$ 
12:  Evaluate  $\mathcal{S}$  and append it to  $\mathcal{A}_{\text{rc}}$ 
13:   $FE \leftarrow FE + |\mathcal{S}|$ 
14:   $\mathcal{P} \leftarrow \text{ENVSELECT}(\mathcal{A}_{\text{rc}}, N)$ 
15: end while
16: return  $\mathcal{A}_{\text{rc}}$ 

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the radial-grid environmental selection of RSEA [3] over the whole archive.

Algorithm 1 states the procedure; its grouping and candidate selection steps are detailed in Algorithms 2 and 3, respectively. The initial design satisfies $2 \leq N_{\text{init}} \leq FE_{\max}$, and N is the target population size for environmental selection. The relation surrogate $\mathcal{M}_R$ includes the group references and input normalisation needed for prediction, while $\mathcal{M}_I$ denotes the indicator surrogate when available. The controls are shared by the three procedures. All expensive evaluations after initialisation are consumed in step (v), so both proposed mechanisms operate on already evaluated data and surrogate predictions only, and neither changes the total number of real evaluations. The two interfaces stay separate: grouping decides *what the classifier is asked to learn*, candidate selection decides *how its scores are spent*.

Figure 1 shows the evaluation loop and expands CDIS into its relation-guided candidate search and criterion-dependent infill branches.

3.2. PBI-Assisted Quality Classification (PAQC)

Relation learning requires training groups that reflect the quality of evaluated solutions. REMO [2] derives its grouping signal from a small set of representatives selected through RSEA’s two-dimensional radial grid [3]. These representatives retain their original objective vectors, but the projected distribution guides their selection. This sparse summary may not fully reflect the directional distribution of the current nondominated solutions in the original many-objective space.

PAQC supplements this representative-based grouping signal with quality information derived directly from the current nondominated solutions. It constructs reference directions from their original objective vectors, bypassing radial-grid representative selection, and uses PBI to assess evaluated solutions along these directions. A stage-dependent fusion combines the directional quality information with the original representative-based labels. The highest-ranked fraction forms the positive training group, and the remaining solutions form its complement. PAQC thus retains REMO’s representative-based sig-

nal while adding another information source for relation-group construction.

Figure 2 provides a geometric overview of these two signals and their roles in grouping.

For PAQC, $\mathcal{P} = \{(\mathbf{x}_i, \mathbf{f}_i)\}_{i=1}^N$ is the current population, with $N = |\mathcal{P}|$, and $\mathbf{z}^* = \min_i \mathbf{f}_i$ the component-wise population minimum, taken as the ideal point.

3.2.1. Quality Assessment Using Nondominated-Derived Directions

The current nondominated solutions provide the additional directional information. Let $\{\mathbf{f}^{(j)}\}_{j=1}^{n_{\text{ND}}}$ denote the objective vectors of the first nondominated front. PAQC normalises these original objective vectors to construct the reference-direction set $\mathcal{V} = \{\mathbf{v}_j\}$:

$$\mathcal{V} = \{\mathbf{f}^{(j)} / \|\mathbf{f}^{(j)}\|_2 \mid j = 1, \dots, n_{\text{ND}}\}. \quad (2)$$

The number and orientations of these directions follow the current nondominated distribution in the original objective space. This construction uses the nondominated vectors directly, without radial-grid representative selection. When the adaptive construction is not applicable, PAQC uses uniformly distributed reference directions as a fallback. This includes $M \leq 3$, $N < 50$, too few nondominated solutions, or a degenerate objective range in the nondominated set.

Each objective vector is associated with the direction of largest cosine similarity, $a(i) = \arg \max_j \mathbf{f}_i^\top \mathbf{v}_j / (\|\mathbf{f}_i\|_2 \|\mathbf{v}_j\|_2)$, and the conventional PBI value [1] along that direction is

$$\begin{aligned} d_{1,i} &= (\mathbf{f}_i - \mathbf{z}^*)^\top \mathbf{v}_{a(i)} / \|\mathbf{v}_{a(i)}\|_2, \\ d_{2,i} &= \|(\mathbf{f}_i - \mathbf{z}^*) - d_{1,i} \mathbf{v}_{a(i)} / \|\mathbf{v}_{a(i)}\|_2\|_2, \\ g_i^{\text{PBI}} &= d_{1,i} + \theta d_{2,i}, \end{aligned} \quad (3)$$

with penalty θ . PAQC maps this PBI value to a bounded quality signal:

$$S_i = \frac{1}{1 + g_i^{\text{PBI}}} \in (0, 1], \quad (4)$$

so a larger S_i indicates a smaller PBI value along the associated direction. This transformation expresses the directional assessment as a bounded signal for fusion with the representative-based label during group construction.

3.2.2. Representative-Based Grouping Signal

PAQC retains the representative-based grouping signal used in REMO [2]. RSEA’s radial-grid selection [3] supplies a small set $\mathcal{R} \subset \mathcal{P}$ of k representative solutions from the current population. These representatives induce the original binary grouping signal. The representatives are evaluated solutions that act as dynamic, population-derived reference points. We set the requested representative count to $k = \lceil 1.5M \rceil$ in the considered experiments; the implementation bounds are given in Section 4.1.

Given $\mathcal{R}$, the binary group label applies the PBI partition of REMO [2] with the representatives as reference points. Every solution is associated with the representative $\mathbf{r}_{b(i)}$ of largest

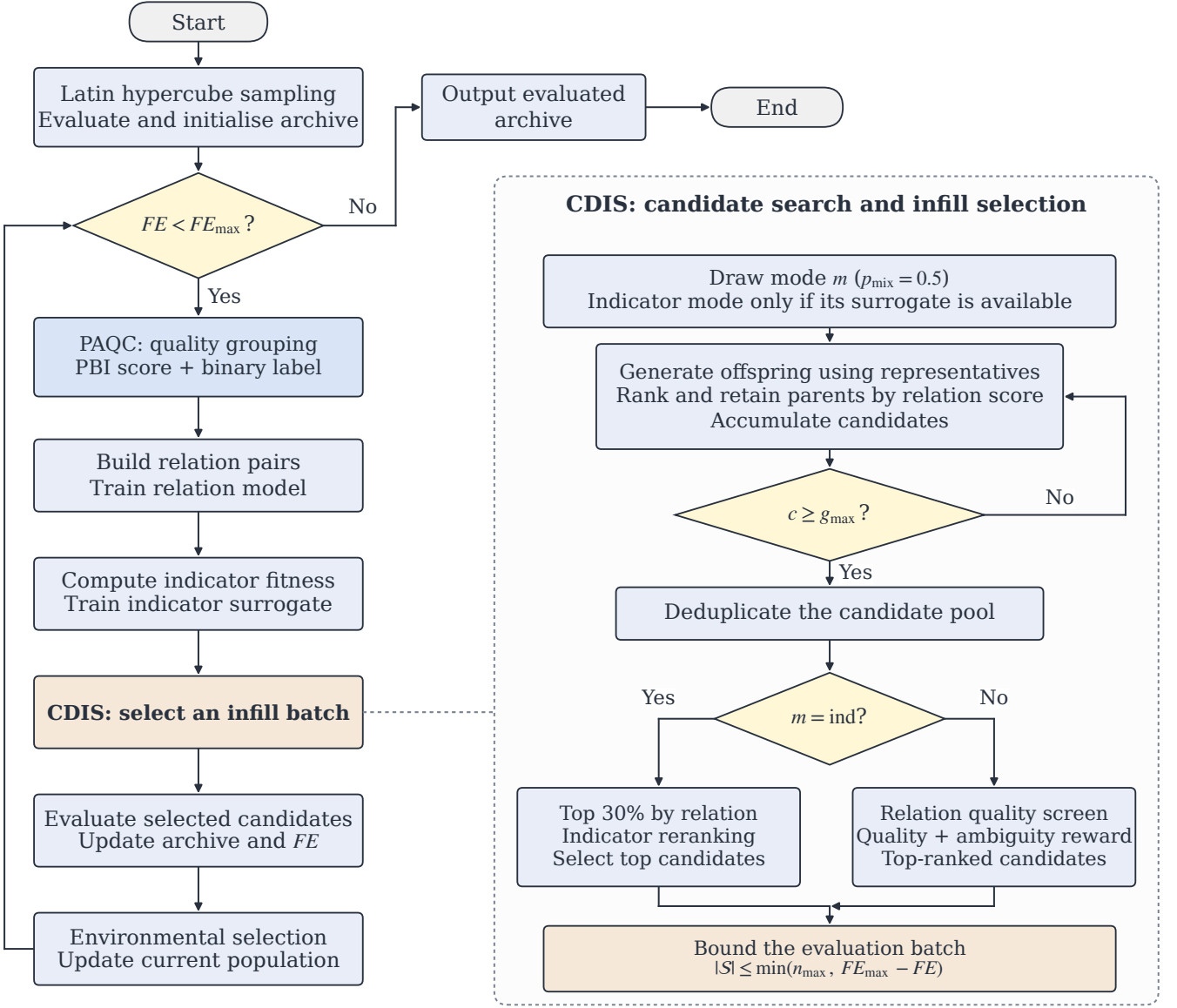


Figure 1: Flowchart of PACDIS. Left: the expensive-evaluation loop, with PAQC constructing the training groups and CDIS selecting the infill batch. Right: expansion of CDIS (dotted connector). A mode is drawn before the relation-guided search; c counts the generated candidates and $g_{\max}$ bounds their accumulation. Indicator mode is drawn with probability $p_{\text{mix}} = 0.5$ when its surrogate is available; otherwise the ambiguity-oriented mode is used. The selected mode determines the final filtering and selection criterion. The ambiguity-oriented mode uses a relation-quality screen followed by ambiguity-rewarded ranking; the indicator branch directly reranks its relation-screened subset. The batch is bounded by its cap and the remaining evaluation budget, without minimum-size completion. Detailed safeguards are given in Algorithms 1 and 3.

cosine similarity, the associated direction is $\mathbf{w}_{b(i)} = (\mathbf{r}_{b(i)} - \mathbf{z}^*) / \|\mathbf{r}_{b(i)} - \mathbf{z}^*\|_2$, the two distances $\hat{d}_{1,i}, \hat{d}_{2,i}$ are formed as in (3) with $\mathbf{v}_{a(i)}$ replaced by $\mathbf{w}_{b(i)}$, and the binary label is

$$L_i = \mathbb{I} \left[\frac{\hat{d}_{1,i} + \delta \hat{d}_{2,i}}{\|\mathbf{r}_{b(i)} - \mathbf{z}^*\|_2} \leq 1 \right] \in \{0, 1\}. \quad (5)$$

Normalising by $\|\mathbf{r}_{b(i)} - \mathbf{z}^*\|_2$ makes the associated representative itself the threshold, so L_i records on which side of the surface through that representative a solution lies. Following REMO, the class-balance variable δ is adjusted by bounded bisection so that the boundary maintains useful occupancy on both sides across changing population states.

3.2.3. Complementary-Signal Fusion for Positive-Group Construction

The representative-based label describes membership relative to the selected reference solutions, while the nondominated-derived directions supply additional quality information from the original objective space. PAQC combines these two information sources with a weight that follows consumption of the expensive budget. With $t = FE / FE_{\max}$ and $\alpha_t = 1 - t$, the hybrid quality score of solution i is

$$H_i = \alpha_t S_i + (1 - \alpha_t) L_i, \quad \alpha_t = 1 - t, \quad (6)$$

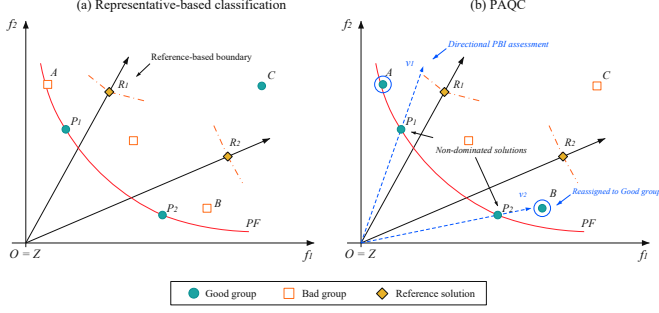


Figure 2: Conceptual illustration of representative-based classification and PAQC. Black rays connect the ideal point Z to representative solutions, and orange dash-dotted segments indicate local classification thresholds. Blue dashed rays show directions constructed from nondominated solutions P_1 and P_2 . In (a), “Good group” and “Bad group” denote the binary labels $L_i = 1$ and $L_i = 0$; in (b), they denote C_1 and C_2 , respectively. Blue rings highlight the reassignment of A and B . The PF is an external reference only. Selected solutions and schematic boundaries are shown, rather than a complete population or a numerically evaluated example. Geometric presentation adapted from REMO [2].

and the positive group collects the highest-ranked fraction r_g of the population,

$$C_1 = \{\mathbf{x}_i \mid \text{rank}_\downarrow(H_i) \leq \lceil r_g N \rceil\}, \quad C_2 = \mathcal{P} \setminus C_1. \quad (7)$$

With $r_g < 1/2$ the non-positive group C_2 provides a broad contrast set, while the compact positive group concentrates pairwise supervision on the highest-priority portion of the population.

Figure 2 illustrates the distinct roles of the two PBI signals. In panel (a), the representatives R_1 and R_2 define the local thresholds used to obtain L_i . In panel (b), the nondominated solutions P_1 and P_2 supply the adaptive directions $\mathbf{v}_1$ and $\mathbf{v}_2$ through (2); these directions provide an additional directional assessment through S_i , rather than directly defining a new classification boundary. The reassignment of A and B to the positive group and of C to its complement schematically depicts how the additional directional information can change selection priorities through (6)–(7). PAQC therefore modifies the ranking used to form the groups, without replacing the representative-based boundary with a new geometric boundary.

The diagram uses two objectives to visualise the two grouping signals; it is not the radial projection used by RSEA and does not demonstrate projection loss. Setting $O = Z$ makes the radial construction in (2) visible as rays through P_1 and P_2 . These two solutions are placed on the PF for illustration, whereas the implementation uses the current nondominated set and does not require the true PF. Their displayed positive-group membership is an illustrative assignment, not a consequence guaranteed by nondominance.

Algorithm 2 summarises the construction. Ties in the hybrid score are resolved by population order so that the positive group has exactly $\lceil r_g N \rceil$ members.

The fusion connects the two information sources to relation learning through the chain

$$\begin{aligned} \text{signals } (S, L) &\rightarrow \text{positive-group construction} \\ &\rightarrow \text{relation-pair construction,} \end{aligned}$$

Algorithm 2 PBIQualityClassification (PAQC)

Input: evaluated population $\mathcal{P}$, progress t

Output: groups C_1, C_2 and representatives $\mathcal{R}$

- 1: Construct $\mathcal{V}$ and associate solutions by cosine similarity
- 2: Compute directional quality signal S_i by (3)–(4)
- 3: Select k representatives $\mathcal{R}$ by radial-grid selection
- 4: Obtain representative-based labels L_i by (5)
- 5: Fuse the two signals: $H_i \leftarrow (1 - t)S_i + tL_i$
- 6: $C_1 \leftarrow$ the top $\lceil r_g |\mathcal{P}| \rceil$ solutions under H_i
- 7: $C_2 \leftarrow \mathcal{P} \setminus C_1$
- 8: **return** $(C_1, C_2, \mathcal{R})$

so a change in the ranking matters insofar as it changes which solutions enter C_1 and therefore which ordered pairs the relation model uses for training. Early in the search, directional quality receives more weight in group construction; as the search consumes the budget, representative-based membership becomes dominant. The group-quality analysis in Section 4.4 will examine whether this weighting improves positive-group quality.

3.3. Relation Learning on the PAQC Groups

PACDIS retains the relation model of REMO [2] and supplies it with the groups constructed by PAQC. Every ordered pair of distinct solutions becomes one training sample whose input is the concatenation $[\mathbf{x}_i, \mathbf{x}_j] \in \mathbb{R}^{2D}$ and whose label

$$y_{ij} = \begin{cases} +1, & \mathbf{x}_i \in C_1, \mathbf{x}_j \in C_2, \\ -1, & \mathbf{x}_i \in C_2, \mathbf{x}_j \in C_1, \\ 0, & \text{otherwise,} \end{cases} \quad (8)$$

with C_1, C_2 from (7), records the ordered group relation rather than a Pareto comparison. Both same-group pair types retain label 0 regardless of their members’ S_i or H_i values. PAQC therefore changes the composition of the training groups; the relation labels do not encode quality ordering within either group. Because $r_g < 1/2$ makes the same-group families larger than the cross-group ones, they are randomly subsampled towards the cross-group count, so the three labels of (8) are balanced by construction rather than by a reweighted loss.

We employ the same relation-learning architecture as REMO: min-max normalised pair inputs, a class-balanced training/held-out split, and a feed-forward pattern network ending in a softmax over the three labels. Its output for a pair is written $\pi(\mathbf{x}_a, \mathbf{x}_b) = [\pi_{+1}, \pi_0, \pi_{-1}]$, the predicted probabilities that the first solution belongs to the higher group, that both belong to the same group, and that the second belongs to the higher group. The held-out partition is retained by the training procedure; no held-out error is used to gate candidate selection in the present implementation.

For an unevaluated candidate $\mathbf{x}$ the trained network is queried on the four families of ordered comparisons $(C_1, \mathbf{x})$, $(\mathbf{x}, C_1)$, $(C_2, \mathbf{x})$ and $(\mathbf{x}, C_2)$, that is on $2N$ pairs per candidate. With $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ the family-wise mean probability vectors in that order, the relation score aggregates the evidence that $\mathbf{x}$ is not inferior to the positive group and is superior to the non-positive group, minus the opposite evidence,

$$R(\mathbf{x}) = 2[c_{-1}(\mathbf{x}) + d_{+1}(\mathbf{x}) - a_{+1}(\mathbf{x}) - b_{-1}(\mathbf{x})] \in [-4, 4], \quad (9)$$

the bound following from each probability vector summing to one. $R(\mathbf{x})$ is the aggregate group-relative preference used to order unevaluated candidates.

3.4. Criterion-Diversified Infill Selection (CDIS)

PAQC determines the supervision supplied to the relation model, whereas CDIS determines how the resulting surrogate information guides expensive-evaluation allocation. CDIS uses two selection criteria: ambiguity-rewarded ranking after relation-quality screening, and indicator-based reranking of a relation-screened subset. The first criterion uses the relation model's confidence information; the second uses a separate indicator surrogate. A fixed cap bounds each expensive-evaluation batch.

Both modes use the same candidate-pool construction procedure. Genetic variation is applied to the current population and representatives $\mathcal{R}$; the relation score in (9) ranks the offspring, and up to $|\mathcal{R}|$ of the best are retained as parents of the next inner generation together with $\mathcal{R}$. The search stops once the cumulative number of generated candidates reaches $g_{\max}$. All generated candidates are accumulated and deduplicated into $\mathcal{A}$. The mode is drawn before this search: mode-specific aggregation of the pair predictions can change the inner rankings and hence the realized pool. Thus, the modes share a pool-construction procedure, rather than necessarily an identical pool.

3.4.1. Ambiguity-Rewarded Infill

The ambiguity-oriented mode first applies a relation-quality screen and then ranks the retained candidates using predicted quality and classification ambiguity. Its relation score uses the largest softmax class probability as the weight of each ordered pair in the family-wise averages of (9). This pair-level confidence weighting reduces the influence of unreliable relation predictions when estimating candidate quality.

Candidates at or above the q_{keep} quantile of the relation score form

$$\mathcal{H}^{\text{amb}} = \{\mathbf{x} \in \mathcal{A} : R(\mathbf{x}) \geq Q_{q_{\text{keep}}}(R(\mathcal{A}))\}. \quad (10)$$

With $q_{\text{keep}} = 0.70$, this screen usually retains the highest-scoring 30% of the pool; ties can increase that fraction. Ambiguity is considered only after this quality screen, so a candidate with poor predicted relation quality cannot enter the evaluation batch solely because of high prediction uncertainty.

To distinguish candidates within the retained set, we additionally measure the ambiguity of the relation predictions. Let $\mathcal{O}(\mathbf{x})$ denote all ordered comparisons constructed between candidate $\mathbf{x}$ and the positive and non-positive training groups, and let $\pi_k(\mathbf{u}, \mathbf{v})$ be the predicted probability of relation class $k \in \{-1, 0, +1\}$ for an ordered pair $(\mathbf{u}, \mathbf{v})$. The candidate-level ambiguity is defined as

$$U(\mathbf{x}) = 1 - \frac{1}{|\mathcal{O}(\mathbf{x})|} \sum_{(\mathbf{u}, \mathbf{v}) \in \mathcal{O}(\mathbf{x})} \max_{k \in \{-1, 0, +1\}} \pi_k(\mathbf{u}, \mathbf{v}). \quad (11)$$

Hence, $U(\mathbf{x})$ becomes large when the relation network is less decisive over the comparisons associated with $\mathbf{x}$.

Within $\mathcal{H}^{\text{amb}}$, the relation score and ambiguity are min-max normalized and combined as

$$A_{\text{amb}}(\mathbf{x}) = \tilde{R}(\mathbf{x}) + \lambda \tilde{U}(\mathbf{x}), \quad \lambda = 0.30, \quad (12)$$

where a constant quantity is assigned the normalized value 0.5. The fixed ambiguity weight λ provides a moderate preference for candidates whose predicted relations remain uncertain while preserving relation quality as the dominant component of the ambiguity-rewarded acquisition. The weight λ remains constant throughout optimization.

The confidence weighting used in $R(\mathbf{x})$ and the ambiguity reward in (12) serve complementary purposes. The former stabilizes the quality estimate by reducing the contribution of uncertain pairwise predictions, whereas the latter uses the remaining candidate-level ambiguity as an exploration signal after the quality requirement has already been satisfied. Thus, uncertainty does not replace the relation-quality criterion but refines the choice among candidates that have passed the quality screen.

The ambiguity-oriented branch sorts $\mathcal{H}^{\text{amb}}$ by descending A_{amb} and returns the first n_b candidates:

$$\mathcal{S} = \text{Top}_{n_b}(\mathcal{H}^{\text{amb}}; A_{\text{amb}}). \quad (13)$$

Here Top_{n_b} returns an ordered list and preserves candidate order for exact score ties. Equation (16) defines the batch size.

The quality screen restricts the eligible set, and the ambiguity reward changes the ordering within that set. Classification ambiguity supplies a selection heuristic; its value alone does not establish objective improvement. The component study must assess the final optimization effect of this rule.

3.4.2. Indicator-Based Infill

The indicator-oriented mode provides an alternative assessment of candidates that pass relation-based screening. The quality indicator combines shift-based density estimation (SDE) [5] with generalized L_p front-shape adaptation [4] and a finite convergence fallback. We compute this indicator for each evaluated solution and train an RBF-SVR model to predict it from decision vectors,

$$\hat{I}(\mathbf{x}) = \mathcal{M}_I(\mathbf{x}; \mathcal{D}), \quad \mathcal{D} = \{(\mathbf{x}_i, I_i)\}_{i=1}^N. \quad (14)$$

The relation score first retains the top $\lceil q_{\text{rel}} |\mathcal{A}| \rceil$ candidates, with $q_{\text{rel}} = 0.30$, to form $\mathcal{H}^{\text{ind}}$. In this mode the family-wise relation probabilities are arithmetic averages. The screened candidates are then ranked by descending $\hat{I}$, and up to $n_{\max}$ are selected directly. Failed or nonfinite indicator predictions fall back to relation-score ranking. This branch separates candidate screening from final quality assessment: relation predictions retain high-scoring candidates, and the indicator surrogate ranks the screened subset.

3.4.3. Selection-Criterion Switching

Let $m \in \{\text{amb}, \text{ind}\}$ denote the mode. With $u \sim U[0, 1)$,

$$m = \begin{cases} \text{ind}, & \text{if the indicator surrogate is available} \\ & \text{and } u < p_{\text{mix}}, \\ \text{amb}, & \text{otherwise,} \end{cases} \quad (15)$$

Algorithm 3 DiversifiedInfillSelection (CDIS)

Input: population $\mathcal{P}$, representatives $\mathcal{R}$, surrogates $\mathcal{M}_R, \mathcal{M}_I$, mode m

Output: ordered candidate batch S

```
1: Run relation-guided variation from  $\mathcal{P} \cup \mathcal{R}$  using mode  $m$ 
2:  $\mathcal{A} \leftarrow$  the accumulated, deduplicated candidate pool
3: if  $m = \text{ind}$  then
4:    $\mathcal{H}^{\text{ind}} \leftarrow$  top  $q_{\text{rel}}$  fraction by  $R$ 
5:   Rerank  $\mathcal{H}^{\text{ind}}$  by indicator predictions  $\hat{I}$ 
6:    $S \leftarrow$  the first  $n_b$  candidates, with  $n_b$  from (16)
7: else
8:    $\mathcal{H}^{\text{amb}} \leftarrow$  quality-screened candidates by (10)
9:   Compute  $A_{\text{amb}} = \bar{R} + 0.30\bar{U}$  within  $\mathcal{H}^{\text{amb}}$ 
10:   $S \leftarrow \text{Top}_{n_b}(\mathcal{H}^{\text{amb}}, A_{\text{amb}})$  by (13)
11: end if
12: return  $S$ 
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where $p_{\text{mix}} = 0.5$. One intact criterion is used per iteration; the two scores need not be combined into a single ranking. This is a fixed probability rule, with the ambiguity-oriented mode as the fallback when the indicator surrogate is unavailable. The feasible batch size is

$$n_b = \min(n_{\text{max}}, |\mathcal{H}^m|, FE_{\text{max}} - FE), \quad (16)$$

with $n_{\text{max}} = 6$. A small retained set produces a smaller batch; it is not completed to a prescribed minimum. If the pool is empty, genetic variation provides fallback candidates, which are deduplicated and truncated to the same cap and remaining budget. Algorithm 3 states the main selection procedure. The framework truncates any returned batch to the remaining budget before its real evaluation and the corresponding FE update.

3.5. Computational Complexity

Let N be the population size, $N_v = O(N)$ the number of directions, N_c the size of the deduplicated candidate pool, D the number of decision variables and M the number of objectives.

In PAQC, association and the continuous PBI computation (3)–(4) cost $O(NN_vM)$, the direction set (2) costs one normalisation of the nondominated vectors, the bounded bisection of (5) costs $O(B_\delta NkM)$ with B_δ fixed by the bracket and the tolerance, and representative selection adds nondominated sorting and radial-projection distances. With $N_v = O(N)$ and $k = O(N)$ the module is dominated by $O(N^2M)$, and generating the relation pairs adds $O(N^2D)$.

In CDIS, scoring one candidate requires $2N$ network inputs of dimension $2D$, so the pool costs $O(N_cND)$ in data construction and prediction, which dominates the module. The indicator fitness costs $O(N^2M)$, the indicator-oriented mode adds one SVR fit on the N evaluated solutions plus predictions on the screened subset. Sorting the ambiguity-rewarded acquisition in (13) costs $O(N_c \log N_c)$. The corresponding cost, excluding surrogate fitting, is $O(N_cND + N^2M + N_c \log N_c)$. Neither module directly consumes expensive evaluations. The remaining-budget check ensures that PACDIS does not exceed FE_{max} , regardless of the mode sequence.

4. Experimental Studies

4.1. Experimental Settings

Test problems and evaluation budget. The main performance study covers DTLZ1–DTLZ7 and WFG1–WFG9 with $M \in \{10, 15, 20\}$, giving 48 problem–objective configurations. These settings target expensive optimization with relatively large numbers of objectives. The main comparison uses PlatEMO [13], with a configured population size of $N = 100$ and a maximum budget of $FE_{\text{max}} = 500$ real function evaluations, including initialization. The decision dimension was $D = 30$, except for WFG2 and WFG3 at $M = 10$ and $M = 20$, where the problem definitions required $D = 31$. All 15-objective instances used $D = 30$.

Compared algorithms. PACDIS is compared with six surrogate-assisted algorithms: REMO [2], SSDE [6], PC-SAEA [7], SAMOEAL2M [8], CSEA [9], and HES-EA [10]. REMO learns pairwise relations, SSDE uses a self-organizing surrogate, and PC-SAEA uses pairwise comparison. SAMOEAL2M uses two-level model management, CSEA uses classification, and HES-EA combines hierarchical and ensemble surrogates with model reduction. The configured population size N does not specify a common initialization count across these implementations.

PACDIS parameters. The same configuration is used across all three objective counts: $g_{\text{max}} = 3000$, $p_{\text{mix}} = 0.50$, $r_g = 0.25$, $q_{\text{keep}} = 0.70$, and $n_{\text{max}} = 6$. The fixed internal constants are the ambiguity weight $\lambda = 0.30$, indicator shortlist fraction $q_{\text{rel}} = 0.30$, and continuous-PBI penalty $\theta = 5$. The requested representative count is

$$k_{\text{eff}} = \min\{N, \max(6, \lceil 1.5M \rceil)\}, \quad (17)$$

giving 15, 23 and 30 at $M = 10, 15$ and 20 with $N = 100$. The actual returned representative count can be smaller. No error-gate threshold or minimum-batch parameter is used in this version.

Implementation details. The implementation retains uniform reference vectors for low-dimensional or small-population cases and when adaptive direction construction is unavailable. For reproducibility, cosine association uses the original objective vectors, whereas PBI distances use vectors translated by $\mathbf{z}^*$, as specified in (3).

Performance indicator and statistical reporting. Performance is measured by inverted generational distance plus (IGD⁺), with smaller values indicating a better approximation to the reference Pareto front. Tables 1 and 2 group the results by problem, with three rows for $M = 10, 15$ and 20 . Each entry reports the mean above the standard deviation in parentheses. The cell with the lowest mean among the seven retained algorithms is bold. The pairwise statistical annotations come from the supplied workbooks. A + indicates that the baseline was reported as better than PACDIS, a – indicates that it was reported as worse, and an = indicates no detected difference. The bottom

row of each table counts these symbols in the order $+/-/=$ for each baseline across all problem-objective configurations in that table. These counts summarize the supplied pairwise outcomes and are not a separate omnibus test across problems.

The tables use available runs with identifiers 1–20. All 48 PACDIS cells contain 20 runs. At $M = 15$, REMO cells contain 10–20 runs; at $M = 20$, the REMO, PC-SAEA, CSEA, and HES-EA cells contain 18–20, 17–20, 19–20, and 17–20 runs, respectively. All other cells contain 20 runs. Thus, the reported means and standard deviations do not all have the same sample size. The export script applies an unpaired rank-sum test at $p < 0.05$ without a multiple-comparison adjustment. We retain its symbols without interpreting them as adjusted significance. We configure a 500-evaluation budget; actual terminal FE can differ across implementations. [TODO: Document the IGD⁺ reference sets, execution environment, and actual terminal FE by algorithm. Audit the rank-sum export when sample sizes differ and specify a multiple-comparison policy for final inference.]

4.2. Comparison with the Six Baseline Algorithms

Tables 1 and 2 report the 500-evaluation comparison. PACDIS has the lowest mean IGD⁺ on 7, 5, and 4 of the 16 problems at $M = 10$, 15, and 20, respectively. These best means occur on DTLZ4, DTLZ6, WFG2, and WFG7 at all three objective counts.

HES-EA has the lowest mean on DTLZ5, DTLZ7, and WFG8 at all three objective counts. Its $+/-/=$ counts against PACDIS are 6/8/2, 5/8/3, and 5/7/4 at $M = 10$, 15, and 20. REMO’s corresponding counts are 1/9/6, 2/9/5, and 3/9/4. SSDE, PC-SAEA, SAMOEA-TL2M, and CSEA each have more reported losses than wins against PACDIS at every objective count. WFG3 remains a weakness: PACDIS has no lowest mean there, and several baselines receive $+$ annotations. These results describe complete-algorithm performance under the selected budget. They do not attribute the differences to PAQC or CDIS individually.

Table 1: Comparison of IGD+ values on DTLZ1–7.

Problem	M	REMO	SSDE	PC-SAEA	SAMOEa-TL2M	CSEA	HES-EA	PACDIS
DTLZ1	10	1.5590e+2 (3.66e+1) = 8.4392e+1	3.0125e+2 (2.19e+1) – 1.9393e+2	3.4691e+2 (2.78e+1) – 2.3930e+2	2.9606e+2 (4.62e+1) – 2.0313e+2	1.5453e+2 (2.45e+1) – 9.6883e+1	2.4889e+2 (2.64e+1) – 1.6956e+2	1.3723e+2 (3.28e+1) 8.8886e+1
	15	(2.49e+1) = 4.0685e+1	(3.10e+1) – 1.1019e+2	(3.42e+1) – 1.3841e+2	(3.17e+1) – 1.2115e+2	(2.07e+1) = 4.2737e+1	(3.01e+1) – 8.0751e+1	(2.40e+1) 4.7041e+1
	20	(1.50e+1) +	(2.33e+1) –	(2.33e+1) –	(2.25e+1) –	(1.23e+1) =	(1.29e+1) –	(9.78e+0)
DTLZ2	10	7.2496e-1 (5.65e-2) =	1.9521e+0 (1.78e-1) –	1.3575e+0 (6.73e-2) –	7.3399e-1 (3.11e-2) –	8.5517e-1 (9.41e-2) –	7.6084e-1 (1.35e-1) =	6.8787e-1 (5.43e-2)
	15	6.8640e-1 (5.91e-2) =	1.5363e+0 (2.00e-1) –	1.1974e+0 (5.64e-2) –	7.8171e-1 (1.81e-2) –	7.7488e-1 (5.06e-2) –	6.8679e-1 (5.28e-2) =	6.8015e-1 (3.63e-2)
	20	7.1476e-1 (5.59e-2) =	1.2435e+0 (9.02e-2) –	1.0366e+0 (3.52e-2) –	8.4644e-1 (2.52e-2) –	7.6517e-1 (5.85e-2) –	6.8671e-1 (3.34e-2) =	7.0057e-1 (2.72e-2)
DTLZ3	10	5.7372e+2 (8.87e+1) = 3.1110e+2	1.2148e+3 (1.27e+2) – 8.6963e+2	1.2800e+3 (8.89e+1) – 8.6289e+2	1.2202e+3 (1.08e+2) – 8.1760e+2	5.7177e+2 (1.18e+2) = 3.2313e+2	1.1258e+3 (1.11e+2) – 6.9556e+2	5.8860e+2 (1.13e+2) 3.6359e+2
	15	(8.82e+1) + 1.3530e+2	(1.04e+2) – 5.2801e+2	(1.07e+2) – 4.9694e+2	(5.68e+1) – 4.3707e+2	(4.93e+1) + 1.5265e+2	(1.06e+2) – 3.9014e+2	(4.70e+1) 1.8445e+2
	20	(3.52e+1) +	(6.89e+1) –	(6.87e+1) –	(1.14e+2) –	(4.15e+1) +	(7.51e+1) –	(3.73e+1)
DTLZ4	10	6.3471e-1 (5.89e-2) =	1.5893e+0 (1.47e-1) –	1.4030e+0 (9.50e-2) –	9.2105e-1 (4.45e-2) –	7.3973e-1 (1.05e-1) –	9.9872e-1 (1.52e-1) –	6.1326e-1 (8.90e-2)
	15	6.2645e-1 (5.52e-2) =	1.3889e+0 (1.27e-1) –	1.2075e+0 (4.95e-2) –	9.7789e-1 (7.82e-2) –	6.7266e-1 (6.36e-2) –	8.7470e-1 (1.04e-1) –	6.1464e-1 (5.05e-2)
	20	5.7162e-1 (4.92e-2) =	1.0861e+0 (6.11e-2) –	9.2705e-1 (2.77e-2) –	8.6698e-1 (6.23e-2) –	5.7127e-1 (4.17e-2) =	6.7142e-1 (6.50e-2) –	5.7121e-1 (4.25e-2)
DTLZ5	10	4.3586e-1 (9.56e-2) =	1.3563e+0 (1.50e-1) –	9.4074e-1 (7.33e-2) –	2.7219e-1 (5.45e-2) +	4.9025e-1 (1.31e-1) –	2.2805e-1 (5.46e-2) +	3.8688e-1 (8.54e-2)
	15	3.4354e-1 (7.96e-2) =	1.0122e+0 (1.10e-1) –	6.5774e-1 (5.77e-2) –	3.0819e-1 (6.61e-2) =	3.2306e-1 (8.29e-2) =	1.3526e-1 (3.51e-2) +	2.8290e-1 (7.01e-2)
	20	1.5795e-1 (5.31e-2) =	6.2211e-1 (1.06e-1) –	3.7681e-1 (3.29e-2) –	2.3709e-1 (3.21e-2) –	1.9278e-1 (5.77e-2) =	7.6285e-2 (1.86e-2) +	1.8301e-1 (6.25e-2)
DTLZ6	10	1.5572e+1 (1.12e+0) –	1.4179e+1 (1.14e+0) –	1.7332e+1 (4.82e-1) –	1.7944e+1 (1.68e-1) –	1.4488e+1 (1.07e+0) –	1.5852e+1 (1.08e+0) –	1.3321e+1 (1.03e+0)
	15	1.0259e+1 (1.22e+0) –	1.0941e+1 (8.36e-1) –	1.3096e+1 (3.74e-1) –	1.3659e+1 (8.80e-2) –	1.0248e+1 (1.39e+0) –	1.1823e+1 (7.84e-1) –	9.1612e+0 (1.01e+0)
	20	6.5759e+0 (1.12e+0) –	6.9243e+0 (8.45e-1) –	8.7917e+0 (3.49e-1) –	9.1041e+0 (2.47e-1) –	6.0725e+0 (1.11e+0) –	7.2056e+0 (5.33e-1) –	5.3861e+0 (9.92e-1)
DTLZ7	10	1.0878e+1 (1.65e+0) –	2.2413e+1 (3.67e+0) –	2.7442e+1 (1.66e+0) –	2.8888e+1 (1.53e+0) –	1.8333e+1 (3.04e+0) –	1.3833e+0 (7.34e-2) +	7.7784e+0 (9.46e-1)
	15	1.6202e+1 (1.86e+0) –	3.1420e+1 (7.06e+0) –	3.9735e+1 (5.56e+0) –	4.0298e+1 (5.32e+0) –	3.3877e+1 (4.03e+0) –	1.8045e+0 (8.05e-2) +	1.3195e+1 (9.62e-1)
	20	1.8492e+1 (2.09e+0) –	3.5717e+1 (6.79e+0) –	4.6790e+1 (5.79e+0) –	3.7550e+1 (1.17e+1) –	4.2512e+1 (8.23e+0) –	2.0150e+0 (5.37e-2) +	1.6368e+1 (1.28e+0)
+ / – / =		3/6/12	0/21/0	0/21/0	1/19/1	2/13/6	6/12/3	—

Each entry shows the mean above the standard deviation in parentheses. Bold cells have the lowest mean in their row. Symbols +, – and = indicate that the baseline is reported as better than, worse than or not significantly different from PACDIS, respectively. The last row totals these symbols in the stated order.

Table 2: Comparison of IGD+ values on WFG1–9.

Problem	M	REMO	SSDE	PC-SAEA	SAMOEa-TL2M	CSEA	HES-EA	PACDIS
WFG1	10	2.8264e+0 (9.88e-2) =	3.2223e+0 (5.23e-2) –	3.1097e+0 (1.78e-2) –	3.1355e+0 (2.70e-2) –	2.8733e+0 (1.18e-1) –	2.9750e+0 (3.86e-2) –	2.7994e+0 (5.68e-2)
	15	3.3090e+0 (1.87e-1) =	3.8604e+0 (5.06e-2) –	3.7391e+0 (3.70e-2) –	3.7415e+0 (4.17e-2) –	3.4373e+0 (1.32e-1) =	3.6709e+0 (4.34e-2) –	3.4259e+0 (1.44e-1)
	20	3.7799e+0 (2.49e-1) =	4.3574e+0 (1.04e-1) –	4.2664e+0 (9.91e-2) –	4.2675e+0 (9.95e-2) –	3.8317e+0 (1.49e-1) =	4.1899e+0 (1.02e-1) –	3.8483e+0 (1.72e-1)
WFG2	10	2.3808e+0 (1.01e+0) –	2.6744e+0 (3.88e-1) –	2.9604e+0 (5.00e-1) –	3.1734e+0 (8.72e-1) –	3.0505e+0 (1.21e+0) –	1.5975e+0 (3.12e-1) –	1.3686e+0 (7.70e-1)
	15	4.4823e+0 (1.18e+0) –	4.0334e+0 (6.17e-1) –	5.2645e+0 (8.98e-1) –	5.5333e+0 (1.05e+0) –	5.2313e+0 (1.54e+0) –	2.4646e+0 (4.56e-1) =	2.3440e+0 (9.08e-1)
	20	7.1411e+0 (1.64e+0) –	5.1419e+0 (8.63e-1) –	6.8322e+0 (1.32e+0) –	7.0516e+0 (1.52e+0) –	7.3820e+0 (1.28e+0) –	3.6120e+0 (5.32e-1) –	2.5149e+0 (7.52e-1)
WFG3	10	1.2630e+0 (1.31e-1) +	1.8001e+0 (1.51e-1) –	1.6116e+0 (2.35e-2) –	8.9820e-1 (6.90e-2) +	1.4058e+0 (9.47e-2) =	1.6331e+0 (3.30e-2) –	1.4387e+0 (1.28e-1)
	15	1.4559e+0 (1.80e-1) +	2.2556e+0 (2.19e-1) –	2.0168e+0 (4.26e-2) –	1.2058e+0 (1.26e-1) +	1.5768e+0 (1.51e-1) +	2.0479e+0 (4.64e-2) –	1.9549e+0 (9.35e-2)
	20	1.5755e+0 (3.13e-1) +	2.5881e+0 (2.04e-1) –	2.3300e+0 (8.52e-2) =	1.6188e+0 (1.29e-1) +	1.7364e+0 (2.09e-1) +	2.3852e+0 (8.27e-2) =	2.3140e+0 (1.35e-1)
WFG4	10	6.6068e+0 (8.10e-1) –	3.4760e+0 (6.78e-1) +	7.6144e+0 (3.22e-1) –	8.2162e+0 (3.72e-1) –	7.3546e+0 (8.78e-1) –	4.1869e+0 (1.36e+0) =	3.9797e+0 (6.63e-1)
	15	1.4189e+1 (1.12e+0) –	8.6492e+0 (1.02e+0) +	1.4706e+1 (6.33e-1) –	1.5548e+1 (5.25e-1) –	1.5555e+1 (1.36e+0) –	1.0537e+1 (1.11e+0) –	9.7411e+0 (1.03e+0)
	20	2.4120e+1 (2.01e+0) –	1.5417e+1 (1.68e+0) +	2.4259e+1 (8.49e-1) –	2.5472e+1 (7.35e-1) –	2.5530e+1 (1.21e+0) –	1.9043e+1 (1.95e+0) =	1.7744e+1 (2.02e+0)
WFG5	10	5.7069e+0 (5.37e-1) –	5.5435e+0 (3.77e-1) –	5.7216e+0 (2.45e-1) –	5.7181e+0 (4.60e-1) –	5.9489e+0 (3.98e-1) –	3.2092e+0 (4.99e-1) +	4.1091e+0 (5.37e-1)
	15	1.2756e+1 (8.39e-1) –	1.1703e+1 (5.71e-1) –	1.1932e+1 (5.71e-1) –	1.2040e+1 (9.74e-1) –	1.3349e+1 (1.16e+0) –	8.2389e+0 (7.23e-1) +	9.1940e+0 (1.07e+0)
	20	2.2009e+1 (1.22e+0) –	2.0304e+1 (8.32e-1) –	2.0644e+1 (6.67e-1) –	2.0564e+1 (8.84e-1) –	2.2993e+1 (1.35e+0) –	1.6319e+1 (7.44e-1) +	1.8122e+1 (1.66e+0)
WFG6	10	5.0640e+0 (6.99e-1) –	5.0029e+0 (5.00e-1) –	6.2082e+0 (2.85e-1) –	6.7279e+0 (2.97e-1) –	6.3834e+0 (6.46e-1) –	2.5016e+0 (3.15e-1) +	2.8635e+0 (3.47e-1)
	15	1.1759e+1 (9.07e-1) –	1.0898e+1 (1.12e+0) –	1.2419e+1 (4.04e-1) –	1.3164e+1 (4.78e-1) –	1.3317e+1 (1.08e+0) –	5.7338e+0 (1.77e+0) =	6.6398e+0 (9.28e-1)
	20	1.9801e+1 (1.36e+0) –	1.8692e+1 (1.62e+0) –	2.1220e+1 (7.58e-1) –	2.2372e+1 (8.50e-1) –	2.3291e+1 (1.13e+0) –	1.3703e+1 (1.95e+0) =	1.4066e+1 (1.05e+0)
WFG7	10	5.3178e+0 (6.97e-1) –	5.3766e+0 (7.74e-1) –	6.3804e+0 (3.96e-1) –	6.7710e+0 (3.67e-1) –	5.8886e+0 (1.03e+0) –	4.0867e+0 (8.70e-1) –	3.1272e+0 (3.66e-1)
	15	1.1619e+1 (1.12e+0) –	1.1610e+1 (1.28e+0) –	1.2920e+1 (3.75e-1) –	1.3704e+1 (5.16e-1) –	1.4066e+1 (1.16e+0) –	1.1182e+1 (6.26e-1) –	7.3747e+0 (7.24e-1)
	20	2.0149e+1 (1.58e+0) –	2.0229e+1 (1.40e+0) –	2.1899e+1 (7.30e-1) –	2.3204e+1 (7.10e-1) –	2.2966e+1 (1.95e+0) –	2.0260e+1 (6.29e-1) –	1.5167e+1 (1.32e+0)
WFG8	10	6.1180e+0 (4.76e-1) –	5.3806e+0 (5.15e-1) –	6.4204e+0 (4.10e-1) –	6.7476e+0 (3.65e-1) –	6.5476e+0 (5.23e-1) –	3.1960e+0 (5.48e-1) +	4.1347e+0 (4.06e-1)
	15	1.2729e+1 (5.38e-1) –	1.0721e+1 (1.39e+0) –	1.2772e+1 (4.89e-1) –	1.3719e+1 (3.99e-1) –	1.4182e+1 (8.81e-1) –	7.3139e+0 (1.56e+0) +	8.8951e+0 (9.43e-1)
	20	2.0979e+1 (1.48e+0) –	1.8572e+1 (1.18e+0) –	2.1410e+1 (8.03e-1) –	2.2670e+1 (5.53e-1) –	2.3598e+1 (1.16e+0) –	1.5495e+1 (8.59e-1) +	1.6080e+1 (5.87e-1)
WFG9	10	6.3795e+0 (7.59e-1) –	6.4126e+0 (4.50e-1) –	6.6634e+0 (2.97e-1) –	5.2242e+0 (7.56e-1) –	7.0544e+0 (7.88e-1) –	3.9947e+0 (5.77e-1) +	4.5897e+0 (4.61e-1)
	15	1.4029e+1 (8.92e-1) –	1.2630e+1 (6.13e-1) –	1.2901e+1 (5.93e-1) –	1.1310e+1 (1.36e+0) –	1.4428e+1 (7.02e-1) –	9.4436e+0 (8.76e-1) +	1.0386e+1 (1.45e+0)
	20	2.2439e+1 (1.36e+0) –	2.1862e+1 (8.86e-1) –	2.2096e+1 (7.32e-1) –	2.0911e+1 (1.30e+0) –	2.3468e+1 (1.58e+0) –	1.8036e+1 (8.98e-1) +	1.9220e+1 (1.67e+0)
+ / – / =		3/21/3	3/24/0	0/26/1	3/24/0	2/22/3	10/11/6	—

Each entry shows the mean above the standard deviation in parentheses. Bold cells have the lowest mean in their row. Symbols +, – and = indicate that the baseline is reported as better than, worse than or not significantly different from PACDIS, respectively. The last row totals these symbols in the stated order.

4.3. Component Ablation of PAQC and CDIS

We assess PAQC and CDIS through two controls of the current PACDIS implementation (Full). The w/o PAQC control uses REMO’s representative-based grouping to form relation-training labels and retains CDIS. The control labeled w/o CDIS retains PAQC and fixes selection to the ambiguity-rewarded relation criterion. It removes the indicator branch and criterion switching while retaining CDIS’s ambiguity criterion. Full and both controls retain the same relation model, environmental selection, batch-size cap, and absence of within-batch distance.

We evaluate DTLZ1–7 and WFG1–9 at $M = 10$ and 20 , with $N = 100$ and a 300-evaluation cap that includes initialization. Each configuration has 20 independent runs. Full and the two controls end at 300 evaluations. We measure the final populations with IGD^+ and compare each control with Full using a two-sided, unpaired Wilcoxon rank-sum test at $p < 0.05$. We make no multiple-comparison adjustment to the reported symbols. We also show REMO with $k = 15$ and 30 at $M = 10$ and 20 as a contextual baseline. Its archived runs can exceed 300 evaluations, so we exclude REMO from module attribution. This study uses a separate budget from the 500-evaluation main comparison.

Tables 3 and 4 report the results. The tests favor Full over w/o CDIS on 12 of 32 configurations and over w/o PAQC on 10 of 32 configurations. They favor the controls on three and two configurations, respectively; the others carry no reported difference. Fixed ambiguity-only selection improves IGD^+ on DTLZ6 at both objective counts and DTLZ7 at $M = 10$. Replacing PAQC grouping improves it on WFG3 at both objective counts. PAQC grouping and CDIS’s added indicator and switching path therefore have problem-dependent effects on final IGD^+ . The w/o CDIS control retains the ambiguity criterion, so this comparison does not measure the removal of all CDIS selection logic. These results neither establish equal effect sizes nor a statistical interaction.

4.4. Good-Group Precision Analysis of PAQC

The GGP study is intended to examine whether adding PBI information from the current nondominated directions produces better positive groups than grouping with the selected representatives alone. This comparison assesses the utility of the additional signal; it does not by itself identify radial-projection loss as the cause of any observed difference. On matched evaluated-population checkpoints, the reference binary label L , continuous score S , and hybrid score H will be examined using a common outcome definition. Ranked positive sets will use the same cardinality, with binary-label ties handled by a specified rule; native reference-group results will be identified separately if their size differs. For a selected positive set C_1 and an independently defined outcome-positive set $\mathcal{Y}^+$, precision is

$$GGP = \frac{|C_1 \cap \mathcal{Y}^+|}{|C_1|}. \quad (18)$$

The outcome must not be defined by the same PBI score used to construct the group. Coverage and stage-wise results will accompany precision so that missing observations and changing

outcome prevalence remain visible. Checkpoints from one run are repeated measurements, not independent runs. Early, middle and late stages will be distinguished to examine how group quality changes as the weights of the two signals vary. If future retention or another trajectory-dependent outcome is used, GGP describes association under that search trajectory; it does not replace the final-performance ablation. [TODO: Specify the outcome and observation horizon, checkpoint schedule, tie rule, sample coverage and run-level aggregation; insert GGP results and uncertainty. Verify version and budget compatibility before reusing historical checkpoints.]

4.5. Evaluated-Candidate Quality Analysis of CDIS

The CDIS mechanism analysis will examine the quality of candidates selected for expensive evaluation, complementing the PAQC analysis of training groups. [TODO: Compare the ambiguity-oriented criterion, the indicator-oriented criterion, and relation-only ranking on matched evaluated-population checkpoints and a common candidate pool, with identical batch caps. Use the fraction of selected candidates retained by the next environmental selection as the primary outcome. Specify how to evaluate the union of selected batches under a separate diagnostic budget, fix the archive and outcome definition, and aggregate paired comparisons at run level. Report coverage and uncertainty. This fixed-pool comparison isolates final selection; the performance ablation also includes mode-dependent candidate generation. No CDIS mechanism results are available in this draft.]

4.6. Effect of Selection-Criterion Switching

Equation (15) sets $p_{\text{mix}} = 0.5$ as the default. Table 5 restores the earlier five-setting IGD experiment for Lambdat030 with batch distance. These historical results do not establish the sensitivity of the current NoBatchDist configuration under IGD^+ . [TODO: Evaluate $p_{\text{mix}} = 0, 0.5$, and 1 with the current PACDIS configuration and IGD^+ under matched problems, seeds, initialization, and evaluation budgets at $M = 10$ and 20 . Specify the problem subset before running the experiment and verify the protocol before reusing default runs. Record indicator-model failures: $p_{\text{mix}} = 1$ still permits the ambiguity-oriented mode when the indicator surrogate is unavailable. Insert current-configuration results before drawing sensitivity conclusions.]

[TODO: Rebuild the convergence figure from the six current baselines with the 500-evaluation budget and document the available run counts. The existing 300-evaluation figure uses the previous baseline set, so we exclude it here.]

5. Conclusion

PACDIS modifies supervision construction and expensive-evaluation allocation in relation-based surrogate-assisted optimization. PAQC combines representative-based grouping with directional quality information obtained from the current non-dominated set to construct positive training groups. CDIS allocates expensive evaluations through ambiguity-oriented and

Table 3: IGD⁺ ablation results on DTLZ1–7 with 10 and 20 objectives ($N = 100$; Full and controls: $FE = 300$).

Problem	M	D	REMO	w/o CDIS	w/o PAQC	Full	
DTLZ1	10	30	2.5633e+2 (3.53e+1) [−]	2.2171e+2 (3.55e+1) ⁼	2.3035e+2 (2.85e+1) ⁼	2.0865e+2 (3.22e+1)	
	20	30	1.0212e+2 (2.04e+1) [−]	7.6982e+1 (1.67e+1) ⁼	8.2264e+1 (1.84e+1) ⁼	7.1581e+1 (1.41e+1)	
DTLZ2	10	30	1.0250e+0 (7.08e−2) [−]	1.0719e+0 (1.11e−1) [−]	9.2078e−1 (7.01e−2) ⁼	9.1663e−1 (1.13e−1)	
	20	30	8.8540e−1 (5.12e−2) [−]	9.1620e−1 (7.12e−2) [−]	8.1112e−1 (4.06e−2) ⁼	8.0861e−1 (3.54e−2)	
DTLZ3	10	30	9.7902e+2 (1.40e+2) [−]	9.0026e+2 (1.69e+2) ⁼	9.3610e+2 (1.56e+2) ⁼	8.8342e+2 (1.19e+2)	
	20	30	3.8543e+2 (8.42e+1) [−]	3.4770e+2 (5.32e+1) [−]	3.7727e+2 (8.53e+1) [−]	2.9643e+2 (7.29e+1)	
DTLZ4	10	30	1.0421e+0 (1.15e−1) [−]	9.5863e−1 (1.22e−1) [−]	8.7885e−1 (1.27e−1) ⁼	8.6629e−1 (1.21e−1)	
	20	30	7.6967e−1 (5.07e−2) [−]	7.1116e−1 (5.17e−2) ⁼	7.5541e−1 (7.89e−2) [−]	7.1116e−1 (6.37e−2)	
DTLZ5	10	30	6.4533e−1 (1.02e−1) [−]	6.2406e−1 (7.10e−2) [−]	5.9268e−1 (1.04e−1) ⁼	5.6802e−1 (7.89e−2)	
	20	30	3.1914e−1 (5.14e−2) [−]	2.5455e−1 (7.92e−2) ⁼	2.9675e−1 (4.15e−2) ⁼	2.7779e−1 (7.46e−2)	
DTLZ6	10	30	1.6046e+1 (5.95e−1) [−]	1.3919e+1 (1.01e+0) ⁺	1.6614e+1 (6.07e−1) [−]	1.5206e+1 (8.52e−1)	
	20	30	7.6458e+0 (4.92e−1) [−]	6.1597e+0 (9.18e−1) ⁺	7.6274e+0 (6.02e−1) [−]	7.0517e+0 (7.40e−1)	
DTLZ7	10	30	2.1214e+1 (2.57e+0) [−]	9.9013e+0 (1.70e+0) ⁺	1.6206e+1 (1.76e+0) [−]	1.2138e+1 (1.75e+0)	
	20	30	4.1534e+1 (6.92e+0) [−]	2.3639e+1 (7.57e+0) ⁼	2.8193e+1 (3.54e+0) [−]	2.2885e+1 (4.26e+0)	
+ / − / =			−	0/14/0	3/5/6	0/6/8	−

Entries give mean (standard deviation); lower IGD⁺ is better. Bold indicates the lowest mean in each row, including ties at the displayed precision. The symbols +, −, and = denote results better than, worse than, and with no detected difference from Full, respectively, under unadjusted rank-sum tests at $p < 0.05$. Full denotes PACDIS; w/o PAQC replaces its grouping, whereas w/o CDIS fixes selection to the ambiguity criterion. REMO uses $k = 15$ for $M = 10$ and $k = 30$ for $M = 20$; its archived runs can exceed 300 evaluations.

Table 4: IGD⁺ ablation results on WFG1–9 with 10 and 20 objectives ($N = 100$; Full and controls: $FE = 300$).

Problem	M	D	REMO	w/o CDIS	w/o PAQC	Full	
WFG1	10	30	2.9781e+0 (4.85e−2) [−]	2.9153e+0 (3.62e−2) ⁼	2.9199e+0 (5.07e−2) ⁼	2.9009e+0 (4.89e−2)	
	20	30	4.2041e+0 (1.07e−1) [−]	4.1445e+0 (1.20e−1) ⁼	4.1273e+0 (1.34e−1) ⁼	4.0996e+0 (1.43e−1)	
WFG2	10	31	2.5176e+0 (1.01e+0) [−]	2.2862e+0 (9.12e−1) [−]	2.3129e+0 (1.05e+0) [−]	1.6939e+0 (6.00e−1)	
	20	31	6.2140e+0 (1.82e+0) [−]	4.7975e+0 (1.25e+0) [−]	5.0796e+0 (2.06e+0) [−]	3.5109e+0 (8.96e−1)	
WFG3	10	31	1.4840e+0 (9.29e−2) ⁺	1.6232e+0 (6.09e−2) ⁼	1.4508e+0 (8.07e−2) ⁺	1.5847e+0 (8.28e−2)	
	20	31	2.0065e+0 (1.78e−1) ⁺	2.4942e+0 (1.35e−1) ⁼	2.1114e+0 (1.19e−1) ⁺	2.4608e+0 (1.25e−1)	
WFG4	10	30	6.2095e+0 (7.93e−1) [−]	5.6407e+0 (6.08e−1) ⁼	5.5901e+0 (6.48e−1) ⁼	5.5410e+0 (6.19e−1)	
	20	30	2.2794e+1 (1.23e+0) [−]	2.2100e+1 (1.33e+0) [−]	2.0912e+1 (1.34e+0) ⁼	2.0734e+1 (1.79e+0)	
WFG5	10	30	5.0770e+0 (6.44e−1) ⁼	5.2289e+0 (6.93e−1) ⁼	5.0793e+0 (7.74e−1) ⁼	5.0006e+0 (6.46e−1)	
	20	30	2.0668e+1 (1.06e+0) ⁼	2.1112e+1 (1.22e+0) [−]	1.9652e+1 (1.35e+0) ⁼	2.0143e+1 (1.41e+0)	
WFG6	10	30	4.4139e+0 (4.67e−1) [−]	3.8874e+0 (4.76e−1) ⁼	3.9016e+0 (6.70e−1) ⁼	3.8208e+0 (3.81e−1)	
	20	30	1.8847e+1 (7.93e−1) [−]	1.6793e+1 (1.14e+0) ⁼	1.7392e+1 (1.40e+0) ⁼	1.6738e+1 (1.03e+0)	
WFG7	10	30	4.6479e+0 (6.53e−1) [−]	4.1704e+0 (3.53e−1) ⁼	4.2405e+0 (7.47e−1) ⁼	3.9411e+0 (5.21e−1)	
	20	30	1.9600e+1 (1.26e+0) [−]	1.8214e+1 (1.03e+0) ⁼	1.8259e+1 (8.23e−1) ⁼	1.7584e+1 (1.36e+0)	
WFG8	10	30	5.2600e+0 (4.22e−1) [−]	4.9093e+0 (2.80e−1) [−]	4.8712e+0 (3.64e−1) [−]	4.5513e+0 (5.02e−1)	
	20	30	2.0127e+1 (9.11e−1) [−]	1.8102e+1 (7.66e−1) ⁼	1.8769e+1 (1.27e+0) [−]	1.8019e+1 (8.20e−1)	
WFG9	10	30	6.3482e+0 (5.35e−1) [−]	6.4647e+0 (5.03e−1) [−]	5.6455e+0 (4.86e−1) ⁼	5.4815e+0 (6.37e−1)	
	20	30	2.2006e+1 (1.25e+0) [−]	2.2623e+1 (1.06e+0) [−]	2.0784e+1 (1.43e+0) ⁼	2.0978e+1 (1.54e+0)	
+ / − / =			−	2/14/2	0/7/11	2/4/12	−

Entries give mean (standard deviation); lower IGD⁺ is better. Bold indicates the lowest mean in each row, including ties at the displayed precision. The symbols +, −, and = denote results better than, worse than, and with no detected difference from Full, respectively, under unadjusted rank-sum tests at $p < 0.05$. Full denotes PACDIS; w/o PAQC replaces its grouping, whereas w/o CDIS fixes selection to the ambiguity criterion. REMO uses $k = 15$ for $M = 10$ and $k = 30$ for $M = 20$; its archived runs can exceed 300 evaluations.

Table 5: Historical Lambdat030 sensitivity to the switching probability p_{mix} at $M = 10$ and $M = 20$ (batch-distance version; IGD).

Problem	D	$p_{\text{mix}} = 0$	0.25	0.50	0.75	1.00
<i>10-objective problems</i>						
DTLZ2	30	$1.4673e+0$ ($1.04e-1$) ⁻	$1.1980e+0$ ($9.10e-2$) ⁻	$1.1101e+0$ ($1.23e-1$)	$1.1174e+0$ ($1.01e-1$) ⁼	$1.0575e+0$ ($1.27e-1$) ⁼
DTLZ4	30	$1.4452e+0$ ($9.75e-2$) ⁻	$1.2556e+0$ ($8.69e-2$) ⁼	$1.2554e+0$ ($1.06e-1$)	$1.2135e+0$ ($7.44e-2$) ⁼	$1.2491e+0$ ($1.11e-1$) ⁼
DTLZ7	30	$7.1374e+0$ ($1.16e+0$) ⁺	$8.6532e+0$ ($1.65e+0$) ⁺	$1.0613e+1$ ($1.51e+0$)	$1.2275e+1$ ($2.38e+0$) ⁻	$1.9427e+1$ ($2.99e+0$) ⁻
WFG1	30	$3.2195e+0$ ($3.59e-2$) ⁼	$3.1908e+0$ ($3.58e-2$) ⁼	$3.1888e+0$ ($5.33e-2$)	$3.1902e+0$ ($6.01e-2$) ⁼	$3.1852e+0$ ($5.40e-2$) ⁼
WFG3	31	$1.7658e+0$ ($5.63e-2$) ⁻	$1.7316e+0$ ($5.24e-2$) ⁼	$1.6872e+0$ ($7.88e-2$)	$1.6955e+0$ ($6.56e-2$) ⁼	$1.6929e+0$ ($7.01e-2$) ⁼
WFG8	30	$5.5866e+0$ ($2.28e-1$) ⁼	$5.7157e+0$ ($1.98e-1$) ⁼	$5.7532e+0$ ($2.64e-1$)	$5.8423e+0$ ($2.77e-1$) ⁼	$5.7964e+0$ ($3.59e-1$) ⁼
Average rank		3.67	3.33	2.33	3.17	2.50
+ / - / = vs. 0.50		1/3/2	1/1/4	-	0/1/5	0/1/5
<i>20-objective problems</i>						
DTLZ2	30	$1.2616e+0$ ($5.18e-2$) ⁻	$1.1619e+0$ ($6.89e-2$) ⁼	$1.1415e+0$ ($6.26e-2$)	$1.1292e+0$ ($5.42e-2$) ⁼	$1.1111e+0$ ($3.51e-2$) ⁼
DTLZ4	30	$1.0742e+0$ ($4.16e-2$) ⁼	$1.0558e+0$ ($5.35e-2$) ⁼	$1.0506e+0$ ($5.17e-2$)	$1.0661e+0$ ($4.34e-2$) ⁼	$1.1148e+0$ ($5.15e-2$) ⁻
DTLZ7	30	$2.1062e+1$ ($8.87e+0$) ⁼	$1.6810e+1$ ($4.17e+0$) ⁼	$1.8298e+1$ ($3.42e+0$)	$2.3106e+1$ ($4.24e+0$) ⁻	$3.2237e+1$ ($7.51e+0$) ⁻
WFG1	30	$5.6088e+0$ ($4.12e-2$) ⁼	$5.6173e+0$ ($6.19e-2$) ⁼	$5.5870e+0$ ($5.53e-2$)	$5.5998e+0$ ($5.47e-2$) ⁼	$5.6154e+0$ ($8.21e-2$) ⁼
WFG3	31	$2.8724e+0$ ($9.74e-2$) ⁻	$2.7862e+0$ ($1.34e-1$) ⁼	$2.7794e+0$ ($9.92e-2$)	$2.7655e+0$ ($9.01e-2$) ⁼	$2.7013e+0$ ($1.39e-1$) ⁼
WFG8	30	$2.0465e+1$ ($9.21e-1$) ⁼	$2.0362e+1$ ($8.23e-1$) ⁼	$2.0926e+1$ ($8.66e-1$)	$2.1141e+1$ ($1.17e+0$) ⁼	$2.1744e+1$ ($1.34e+0$) ⁼
Average rank		3.67	2.83	2.17	2.83	3.50
+ / - / = vs. 0.50		0/2/4	0/0/6	-	0/1/5	0/2/4
Overall average rank		3.67	3.08	2.25	3.00	3.00
Overall + / - / = vs. 0.50		1/5/6	1/1/10	-	0/2/10	0/3/9

Smaller IGD is better; bold highlights the lowest mean in each row. The supplied PlatEMO export marks each non-default setting as better than (+), worse than (-), or without a detected difference from (=) $p_{\text{mix}} = 0.5$. We compute average ranks from the mean IGD values within each objective setting.

indicator-oriented candidate-selection criteria after relation-quality screening. The reported ablation comparisons suggest that both components benefit the complete algorithm on different problems, with no uniform component dominance. These results support treating PAQC and CDIS as complementary mechanisms at the two interfaces.

[TODO: Complete the statistical and provenance audits before writing the performance conclusions. Add the matched group-quality and candidate-quality findings when available, and distinguish these mechanism diagnostics from final IGD⁺ performance.]

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